

BCS303 Operating Systems - Dec 2024

5 questions · 40 marks · answered from your notes on 18 Sep 2026

Q1a. [8 marks]

Define an operating system. Explain its role as a resource allocator and as a control program

Definition: An operating system is system software that acts as an intermediary between a user of a computer and the computer hardware.

Role as a Resource Allocator:

- Manages hardware and software resources
- Assigns resources to requesting programs depending on priority
- Examples:

+ CPU time, memory space, file-storage space, I/O devices, shared files are allocated to programs

+ Conflicting requests for resources by different programs are resolved

Role as a Control Program:

- Manages the execution of user programs to prevent errors and improper use of the computer
- Examples:

+ Scheduling processes on the CPU

+ Creating and deleting processes

+ Suspending and resuming processes

+ Providing mechanisms for process synchronization and communication

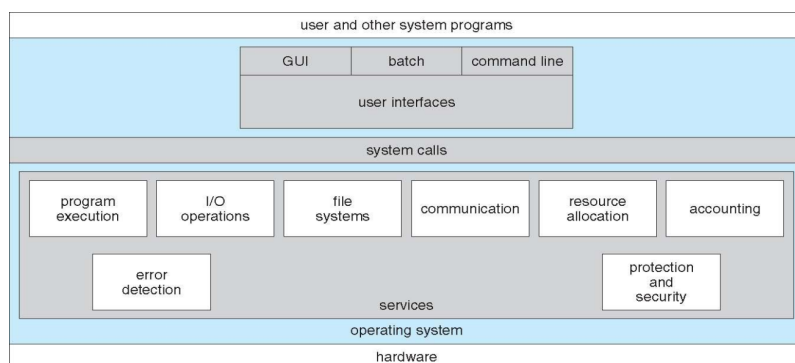


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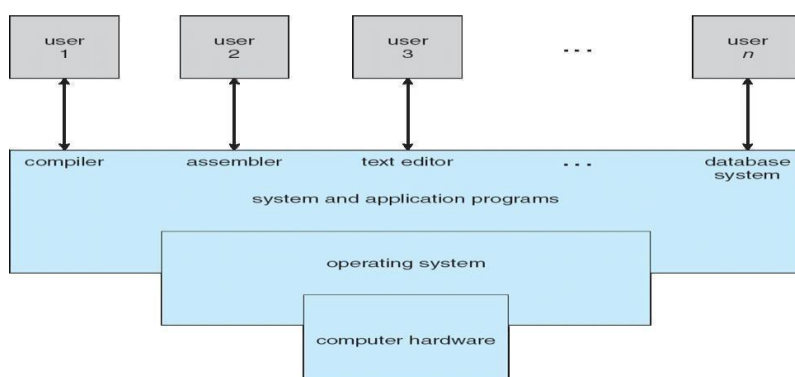


Diagram — page 2

- Process control
 - end, abort
 - load, execute
 - create process, terminate process
 - get process attributes, set process attributes
 - wait for time
 - wait event, signal event
 - allocate and free memory
- File management
 - create file, delete file
 - open, close
 - read, write, reposition
 - get file attributes, set file attributes
- Device management
 - request device, release device
 - read, write, reposition
 - get device attributes, set device attributes
 - logically attach or detach devices
- Information maintenance
 - get time or date, set time or date
 - get system data, set system data
 - get process, file, or device attributes
 - set process, file, or device attributes
- Communications
 - create, delete communication connection
 - send, receive messages
 - transfer status information
 - attach or detach remote devices

Figure: Types of system calls

From your notes: BCS303 module 1, p. 2

Q1b. [7 marks]

With a neat diagram, explain dual-mode operation in an operating system

Dual-Mode Operation

A dual-mode operation is a mode of execution in an operating system where the system can be divided into two separate modes: **User Mode** and **Kernel Mode (Supervisor Mode)**.

- **User Mode:** This mode is used to execute user applications, where the system is not in control. The hardware bit indicating the current mode is set to 0 (user mode).
- **Kernel Mode (Supervisor Mode):** This mode is used by the operating system to manage and control the system resources. The hardware bit indicating the current mode is set to 1 (kernel mode).

Key Points

- The mode bit is used to distinguish between user applications and the operating system.
- When a user application requests a service from the operating system, the transition from user to kernel mode takes place.
- At system boot time, the hardware starts in kernel mode, and the operating system is loaded and starts user applications in user mode.
- Whenever a trap or interrupt occurs, the hardware switches from user mode to kernel mode.
- The dual-mode operation provides protection for the operating system from errant users and prevents errors caused by user programs.

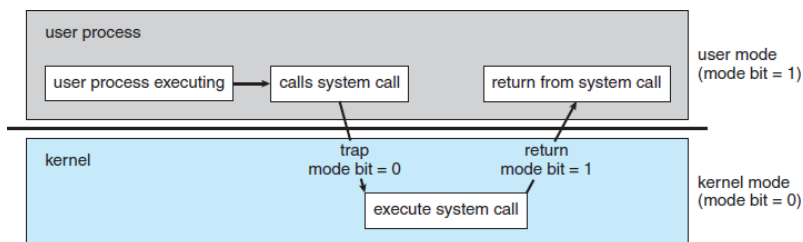


Figure Transition from user to kernel mode.

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From your notes: BCS303 module 1, p. 9

Q1c. [5 marks]

List and explain any five services provided by an operating system

Not answered: ConnectionError(Cannot connect to host opensearch:9200 ssl:default [Temporary failure in name resolution]) caused by: ClientConnectorDNSError(Cannot connect to host opensearch:9200 ssl:default [Temporary failure in name resolution])

Q2a. [10 marks]

Explain the different types of system calls with examples

System Calls

- Definition: System calls provide an interface to the services of the operating system.

Types of System Calls:

- **Hardware System Calls:** These are initiated by hardware interrupts and allow direct access to hardware resources.
- **Software System Calls:** These are initiated by software and allow for interaction with the operating system's services.

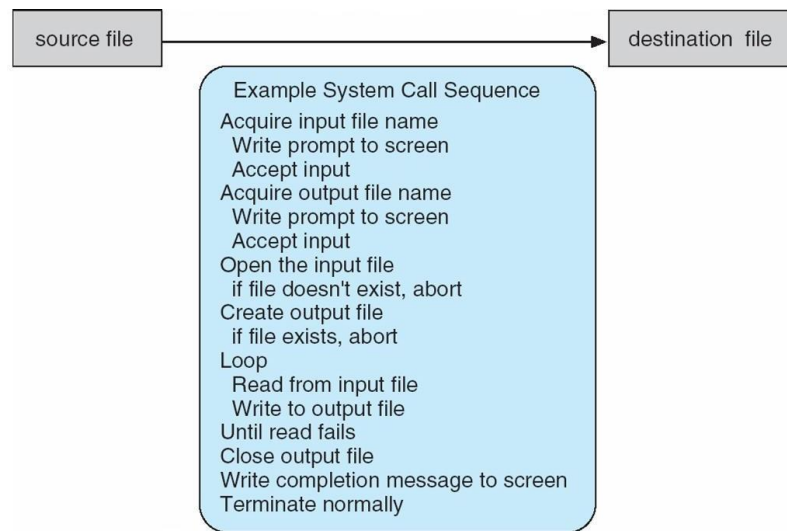


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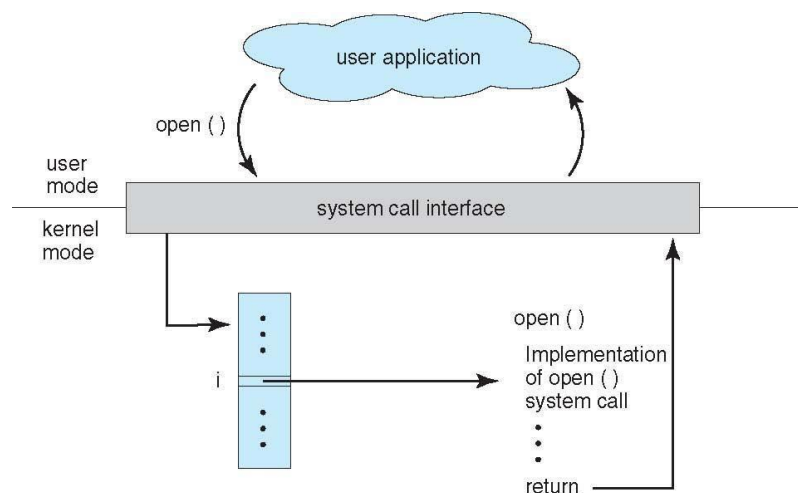


Figure: The handling of a user application invoking the open() system call.

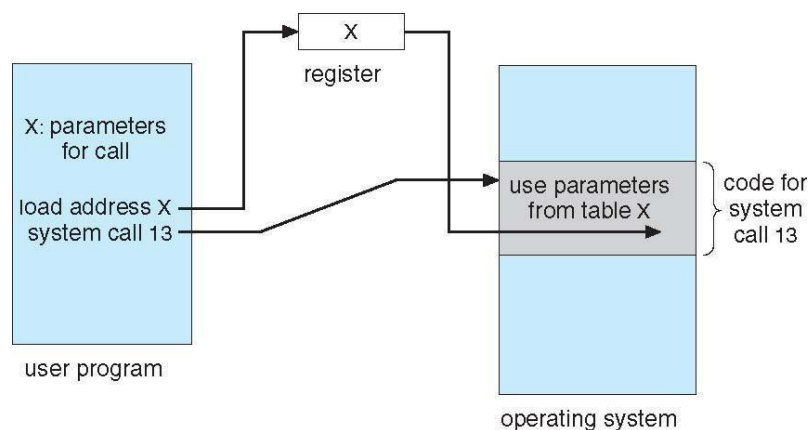


Figure: Passing of parameters as a table.

From your notes: BCS303 module 1, p. 12

Q2b. [10 marks]

Differentiate between multiprogramming and time-sharing systems

Multiprogramming vs Time-Sharing Systems

Definition:

- **Multiprogramming:** A single user cannot keep either the CPU or I/O devices busy at all times. Multiprogramming increases CPU utilization by organizing jobs, so that the CPU always has one to execute.
- **Time-Sharing (Multitasking):** A single CPU executes multiple jobs by switching among them frequently enough that users can interact with each program while it is running.

Key Points:

- **Multiprogramming Systems:**
 - The operating system keeps several jobs in memory simultaneously.
 - The operating system picks and begins to execute one of the jobs in memory.
 - Eventually, the job may have to wait for some tasks, such as an I/O operation, to complete.
 - Multiprogrammed systems provide an environment in which the various system resources are utilized effectively but do not provide for user interaction with the computer system.
- **Time-Sharing (Multitasking) Systems:**
 - A single CPU executes multiple jobs by switching among them frequently enough that users can interact with each program while it is running.
 - Time sharing requires an interactive computer system, which provides direct communication between the user and the system.
 - The response time should be short — typically less than one second.
 - A time-shared operating system allows many users to share the computer simultaneously.

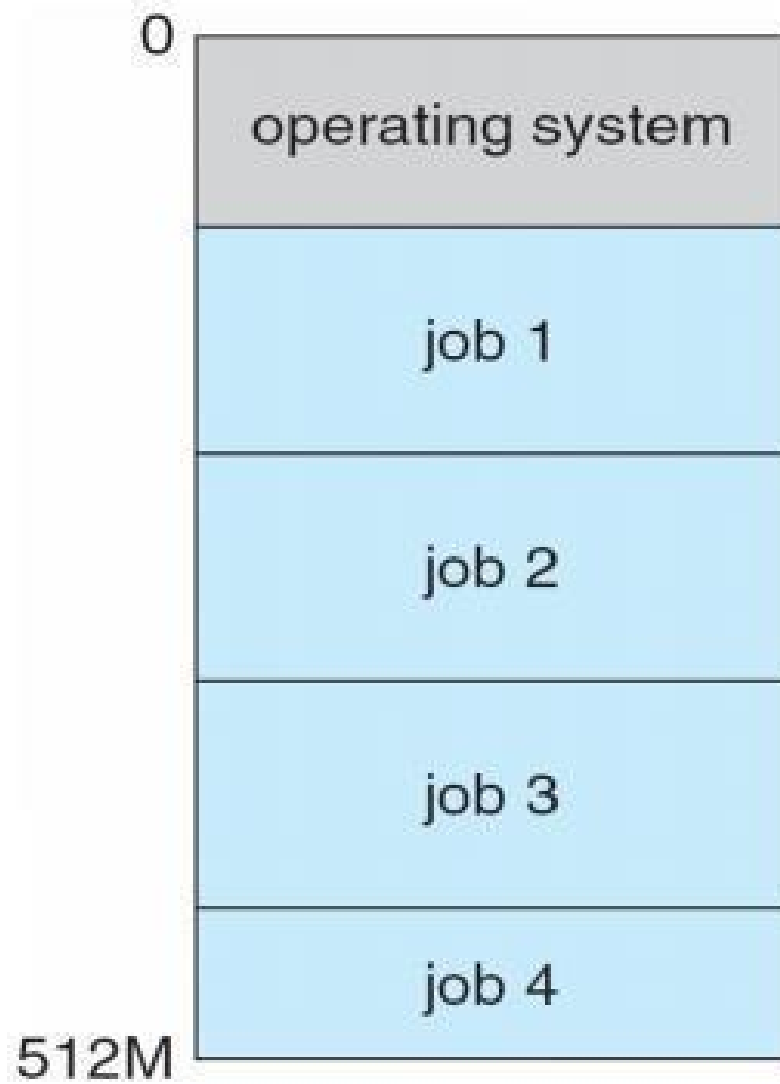


Fig - Memory layout for a multiprogramming system

From your notes: BCS303 module 1, p. 8
